Binary Search

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The problem

- Return the index of target in a sorted array, or -1.

Intuition

- In a sorted array, comparing with the middle element tells us which half can contain the target.

Approach

- 1. Keep left and right bounds
- 2. Check the middle
- 3. Discard the half that can't contain the target
- 4. Return the index or minus one

Java solution

```java
public class BinarySearch {
    public static int binarySearch(int[] array, int target) {
        int lo = 0, hi = array.length - 1;
        while (lo <= hi) {
            int mid = (lo + hi) >>> 1;
            if (array[mid] == target) return mid;
            if (array[mid] < target) lo = mid + 1; else hi = mid - 1;
        }
        return -1;
    }
}
```

Complexity

- Time: O(log n)
- Space: O(1)

Recap

- In a sorted array, comparing with the middle element tells us which half can contain the target.
- Complexity: O(log n)